

AI-Enhanced Intrusion Detection System

Abstract

The AI-Enhanced Intrusion Detection System is a machine learning-based cybersecurity solution designed to identify and classify malicious network traffic. The project uses the CICIDS2017 dataset to detect Distributed Denial-of-Service (DDoS) attacks and distinguish them from normal network traffic. A Random Forest classifier was implemented to achieve high detection accuracy and provide an interactive dashboard for visualizing network security insights.

Problem Statement

Traditional rule-based intrusion detection systems struggle to identify evolving cyber threats and generate high false-positive rates. There is a need for intelligent systems capable of learning traffic patterns and detecting malicious activities with greater accuracy.

Objectives

- Develop an AI-based intrusion detection system.
- Detect DDoS attacks in network traffic.
- Improve threat detection accuracy using machine learning.
- Visualize network security metrics through an interactive dashboard.

Dataset

The project uses the CICIDS2017 dataset, which contains realistic network traffic data and multiple attack scenarios.

Dataset Used: Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

Classes:

- BENIGN
- DDoS

Technology Stack

- Python 3.12
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Streamlit
- Joblib
- GitHub

Methodology

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis
4. Feature Selection
5. Model Training
6. Model Evaluation
7. Dashboard Development

System Workflow

Dataset → Data Cleaning → Feature Extraction → Random Forest Model → Prediction → Dashboard Visualization

Data Preprocessing

- Removed missing values
- Removed duplicate records
- Replaced infinite values
- Standardized column names

Model Development

Algorithm: Random Forest Classifier

Training Samples: 178,465

Testing Samples: 44,617

Results

- Accuracy: 99.99%
- Precision: 1.00
- Recall: 1.00
- F1-Score: 1.00

Feature Importance

Top features influencing intrusion detection:

- Fwd Packet Length Max
- Fwd Packet Length Mean
- Init_Win_bytes_forward
- Bwd Packet Length Min

- Avg Fwd Segment Size

Dashboard Features

- Dataset overview
- Traffic summary
- Attack distribution
- Model performance metrics
- Interactive data preview

Conclusion

The proposed AI-Enhanced Intrusion Detection System successfully detects DDoS attacks with high accuracy using machine learning techniques. The Streamlit dashboard provides an effective interface for monitoring network security metrics and analyzing attack patterns.